HU5175 Education, Technology, and Society

Assignment 1 - Learning Theories

Name	Yuvaraj K
Register Number	2022506079
Branch and Department	B.Tech Information Technology

Table of Contents

S. No.	Title	Page No.
1	Introduction to Learning Theories	1
2	Behaviourism	3
3	Cognitivism	5
4	Social Constructivism	7
5	Humanism	10
6	Additional Learning Theories	13
7	Applying Learning Theories	17
8	References	21

Introduction to Learning Theories

Everybody is a genius. But if you judge a fish by its ability to climb a tree, it will spend its whole life believing that it is stupid."

– Albert Einstein

Learning is generally defined as a continuous and natural process to which the human being is exposed every day of his/her life. This concept is defined by the lasting acquisition of knowledge, behaviours, skills, or abilities through practice, study, or experience.

In this process of learning, there is one very important thing to note: no two humans, or rather, no two students are alike, and the way every person learns will vary. Our brains are all unique, and our various experiences contribute to the different ways we learn. Current and aspiring teachers need to be educated to be prepared for teaching students every day, and an important part of this teacher education is having a concrete understanding of the different learning theories.

When talking about a theory, we must do so, considering they arise as answers to the questions asked by researchers. So, a learning theory is defined as a set of different concepts that observe, describe, explain, and guide the learning process of people and everything that relates to this process.

In simple terms, learning theories are ideas about the ways students learn and retain information. These principles provide different frameworks that teachers can use to adapt to students' diverse learning styles and

academic needs. In addition to helping students absorb the information being taught, learning theories can also help teachers manage students' behaviour. This empowers educators to create an atmosphere that's more inclusive and conducive to learning — a skill that's essential whether teaching online or in a traditional classroom setting.

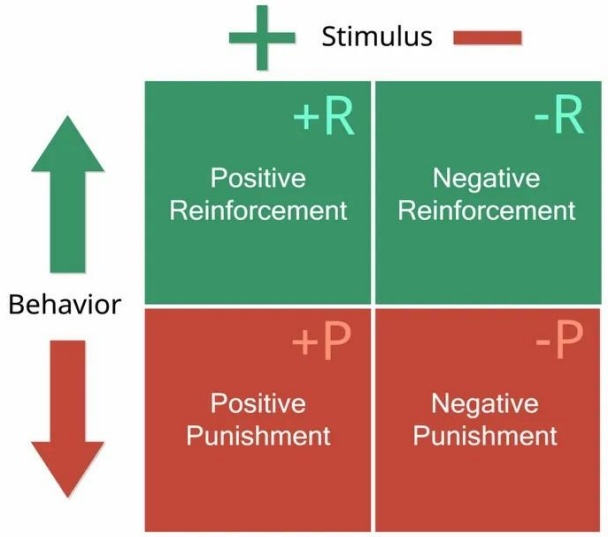
Learning theory describes how students receive, process, and retain knowledge during learning. Cognitive, emotional, and environmental influences, as well as prior experience, all play a part in how understanding, or a worldview, is acquired or changed, and knowledge and skills retained.

There are many solidified learning theories that teachers can learn from as they prepare to help students in the classroom. Teachers who understand learning theories can use different techniques in their classrooms to cater to different kinds of learning. This can help all students find success in learning.

Behaviourism

Founded by **John B. Watson** (but widely associated with **Ivan Pavlov** and **B.F. Skinner**), Behaviorism is the idea that children learn by observing the behaviour of others, whether adults and authority figures or friends and peers their own age.

In Behaviorism, the learner's mind is a *blank slate* ready to absorb knowledge — and **repetition** and **reinforcement** play a key role in communicating with students.



Positive reinforcement is a popular element of behaviourism—classical conditioning observed in Pavlov's dog experiments suggests that behaviours are directly motivated by the reward that can be obtained. Teachers in a classroom can utilise positive reinforcement to help students better learn a concept. Students who receive positive reinforcement are more likely to retain information moving forward, a direct result of the behaviourism theory.

According to Skinner, rewards increase the likelihood that behaviours will be repeated, while punishments decrease the likelihood of repetition. He also theorised that rewards and punishments could be either positive or negative in nature. This can confuse a lot of people! What he meant was that when we give or add something to the environment, the interaction is positive; when we take something away, the interaction is negative. So, for example, removing an undesirable activity from the agenda might be a negative reward, and adding an undesirable activity to it might be a positive punishment.

While behaviourism has contributed valuable insights into learning and behaviour modification, it has been criticised for oversimplifying human learning by ignoring internal cognitive processes and emotions. Critics argue that focusing solely on observable behaviours neglects the complex mental aspects of learning, such as problem-solving, critical thinking, and creativity.

Scenes from a Classroom

Positive Reward or Reinforcement	In Ms. X's first grade classroom, she is working with one of her students, Sam, to help him with some challenging attention-span issues. She gives him a sticker whenever she sees him on task.
Positive Punishment	In Mr. X's third grade classroom, Mr. X quietly reminds a student to keep all four legs of his chair on the floor. Yes, verbal praise & reminders do qualify as adding something to the environment.
Negative Reward or Reinforcement	To increase the number of students who turn in their homework each day, Mr. X announces that if the class has 100% completion on today's assignment, they won't have homework on Friday.
Negative Punishment	Students in Mrs. X's art class are having trouble sharing the supplies. Mrs. X writes the word ART on the board, and each time students have a disagreement over supplies, she erases a letter. If the word is erased completely, students will have to use pencil to complete their project for the day.

Cognitivism

"People say a lot, so I watch what they do."

Introduced during the middle of the 20th century, Cognitivism shifts away from Behaviourism to place a heavier emphasis on the *internal thoughts of the observer*, as opposed to merely observing others' behaviour and responding to stimuli. In fact, this theory emerged in response to the limitations of behaviourism, which didn't adequately explain complex cognitive processes.

Cognitive learning theory looks at the way people think. Mental processes are an important part in understanding how we learn. The cognitive theory understands that learners can be influenced by both internal and external elements. In contrast to Behaviorism, Cognitivism holds that learning chiefly *takes place while the student is working to break down and organise new information in their mind.*

Plato and **Descartes** are two of the first philosophers that focused on cognition and how we as human beings think. Many other researchers looked deeper into the idea of how we think, spurring more research. **Jean Piaget** is a highly important figure in the field of cognitive psychology, and his work focuses on environments and internal structures and how they impact learning.

Cognitive theory has developed over time, breaking off into sub-theories that focus on unique elements of learning and understanding. At the most basic level, cognitive theory suggests that internal thoughts and external forces are both an important part of the cognitive process. And as students understand how their thinking impacts their learning and behaviour, they are able to have more control over it.

This theory impacts students because their understanding of their thought process can help them learn. Teachers can give students opportunities ***to ask questions, to fail, and think out loud***. These strategies can help students understand how their thought process works, and utilise this knowledge to construct better learning opportunities.

Cognitivism has greatly enriched our understanding of learning by focusing on the complexities of mental processes. However, it has been criticised for sometimes *neglecting the influence of social and cultural factors on learning*. Additionally, critics argue that cognitivism *can sometimes overlook the emotional and affective aspects of learning*.

In summary, cognitivism is a learning theory that places mental processes at the centre of the learning experience. It emphasises the role of information processing, schema development, metacognition, and problem-solving in how individuals acquire, organise, and use knowledge. This theory has contributed significantly to shaping modern education and instructional design practices.

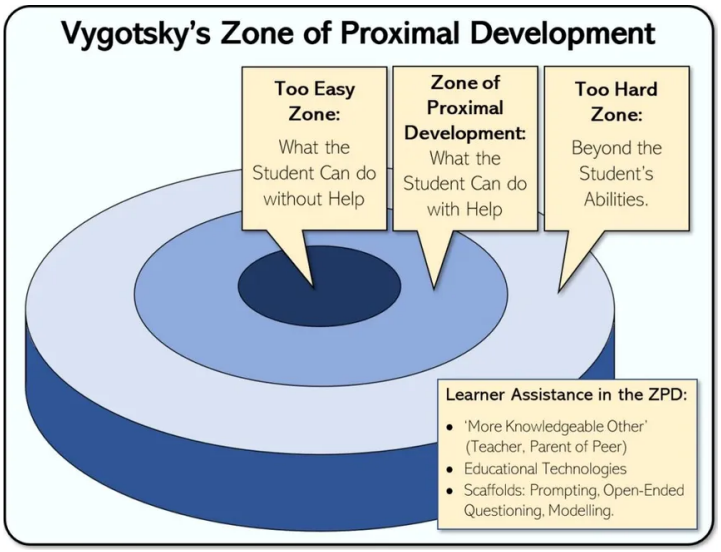
Scenes from a Classroom

To help students memorise the Periodic Table, Mrs. X co-creates various mnemonic devices with her class.
Ms. X uses a graphic organiser in order to help students write paragraphs with appropriate structure.
At the start of every lesson, Mr. X asks questions to activate the prior knowledge of his students. He knows this will help to link the new learning concepts to previously retained ideas, increasing the likelihood that the new learning will be remembered.

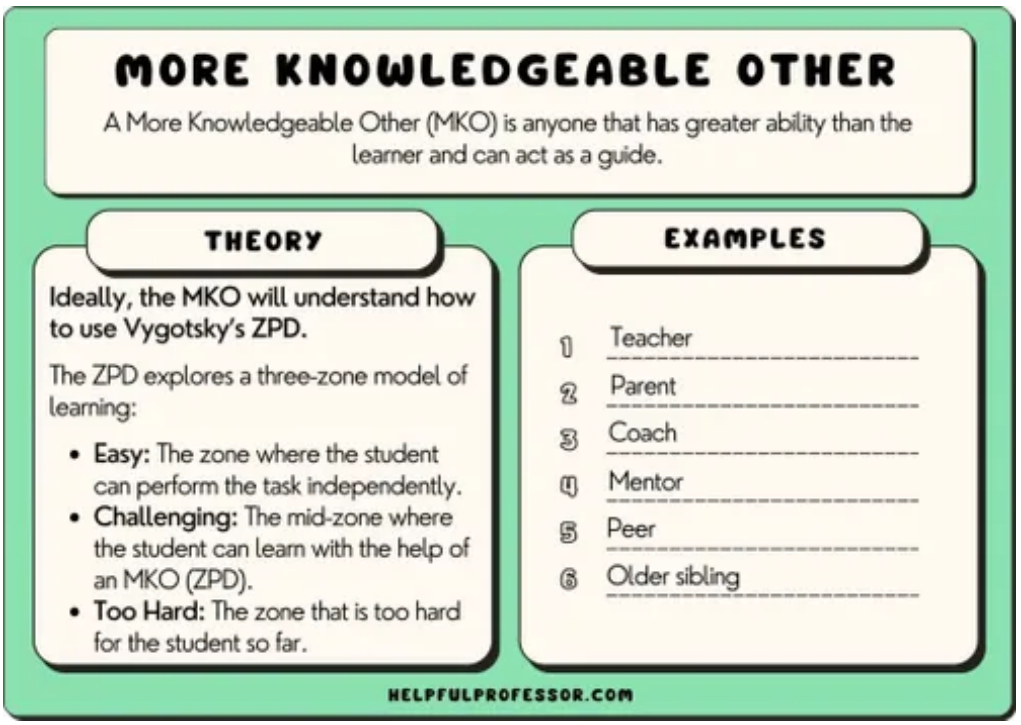
Social Constructivism

Social Constructivism, or simply Constructivism, is a learning theory that is based on the idea that students ***create their own learning*** based on previous experiences. Students take what they're being taught and add it to their previous knowledge and experiences, creating a reality that's unique to them. This theory focuses on learning as something *personal and individual for each student.* It represents a more active approach to learning, as opposed to Behaviorism, where students take a more passive role in learning.

Lev Vygotsky is an important founder of Constructivism. Vygotsky believed that learning is a collaborative process, and that social interaction is fundamental for cognitive development. According to Vygotsky, students learn best when working collaboratively with those whose proficiency level is higher than their own, allowing them to complete tasks they are not yet able to do independently. Vygotsky identified these concepts as the *More Knowledgeable Other (MKO)* and the *Zone of Proximal Development* (ZPD). Constructivist classrooms are student-centred, with the teacher acting as the facilitator.



The Zone of Proximal Development is a concept that refers to the range of tasks that learners can perform with the guidance of a more knowledgeable individual. The Zone of Proximal Development acknowledges that learners can achieve more when assisted by someone who has a deeper understanding of the subject matter; this someone is referred to as the More Knowledgeable Other. There are zones within the Zone of Proximal Development, and zones even beyond it, the latter of which comprises areas where even with the guidance of a more knowledgeable individual, a certain task is very difficult or nearly impossible to successfully complete.



Some other important qualities of constructivism are

- ➢ scaffolding, or temporary support, which is provided to learners as they tackle complex tasks. This support gradually decreases as learners become more competent and confident in their abilities.

- ➢ reflective thinking, where earners are encouraged to reflect on their experiences, identify misconceptions, and adapt their mental models based on new information and insights.

- ➢ learning experiences that mirror real-life situations, which provide a richer environment for constructing knowledge.

Teachers can utilise constructivism to help understand that each student will bring their own past to the classroom. They help students to create their own learning and understanding, their own process and reality based on their past. This is crucial to helping many kinds of students take their own experiences and include them in their learning.

Constructivism offers a powerful framework for understanding learning, but it can downplay the role of explicit instruction and the importance of foundational knowledge. Learners might construct misconceptions or incomplete understanding if not guided effectively.

Scenes from a Classroom

Mr. X pairs students performing on or above grade level with students performing below grade level, inviting them to turn and talk about their learning throughout a lesson.
Mrs. X uses collaborative learning to facilitate engagement with specific learning targets, ensuring heterogeneous student groupings.
Ms. X aids her students in solving real world problems, meeting several learning targets while giving students autonomy to make decisions. She encourages students to work with peers who have different strengths than their own.

Humanism

Founded by pioneers like **Carl Rogers**, **James F. T. Bugental**, and **Abraham Maslow**, The Humanist Teaching and Learning Theory, or simply Humanism, is a learner-centric approach to education that believes in the teaching of the child "as a whole".

Maslow's hierarchy of needs

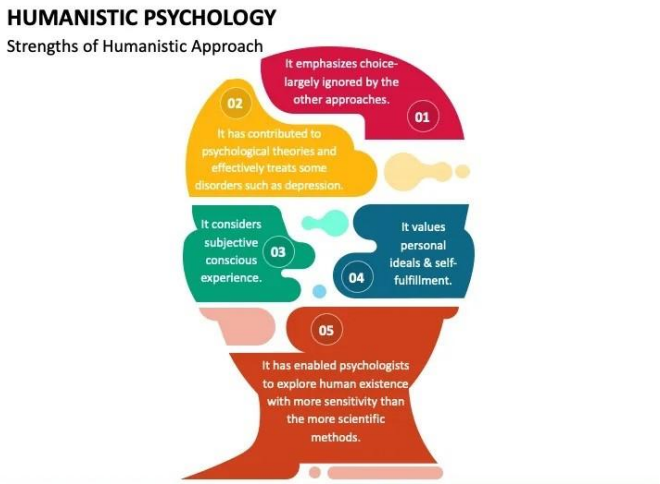
Humanism approaches learning as a way to fulfil an individual's potential rather than meeting specific learning targets. Maslow's research on the ***Hierarchy of Needs*** is a major concept within this theory, as it focuses on the whole person, specifically the cognitive and affective needs of the learner.

This theory places a heavier emphasis on the learner themselves, and their untapped potential, rather than the methods of learning or the materials being taught. Built on the premise that humans are fundamentally good and will act appropriately if their basic needs are met, Humanism prioritises meeting the unique

emotional and academic needs of each learner so that they are empowered to take greater control over their own education.

Humanism is very closely related to constructivism. It directly focuses on the idea of *self-actualization*. Everyone functions under a hierarchy of needs. Self-actualization is at the top of the hierarchy of needs—it's the brief moments where a person feels all of their needs are met and that they're the best possible version of themselves. Everyone is striving for this, and learning environments can either move toward meeting needs or away from meeting needs.

Feelings and knowledge are both important to the learning process and should not be separated according to humanistic psychology. Humanistic teachers believe that knowledge and feelings go hand-in-hand in the learning process. Cognitive and affective learning are both important to humanistic learning. Lessons and activities should focus on the whole student and their intellect and feelings, not one or the other.



A safe learning environment is also crucial in humanism. Educators understand that they need to create a safe environment so that students can have as many as their needs met as possible. They need to feel safe physically, mentally, and emotionally in order to be able to focus on learning. So humanistic educators are passionate about the idea of helping students meet as many of their needs as possible.

A very crucial idea to this theory is choice. Humanistic learning is student-centred, so students are encouraged to take control over their education. They make choices that can range from daily activities to future goals. Students are encouraged to focus on a

specific subject area of interest for a reasonable amount of time that they choose. Teachers who utilise humanistic learning believe that it's crucial for students to find motivation and engagement in their learning, and that is more likely to happen when students are choosing to learn about something that they really want to know.

Scenes from a Classroom

Ms. X begins each day with a morning meeting to check in on her students' emotional well-being and proactively teach them specific coping skills & strategies.
Mr. X, an 8th grade science teacher, provides his students a menu of assessment options to illustrate their mastery of learning targets for the unit.
Mrs. X, a 2nd grade teacher, invites each student to set their reading goal for the quarter.
Mr. X, the school's counsellor, partners with local organisations to fill backpacks with food that students can take home to ensure they have food to eat over the weekend.
Mrs. X, a 5th grade teacher, sets aside an hour of time each week for students to learn about and create anything they want, utilising the framework she provides.

Additional Learning Theories

While the aforementioned theories are the main educational learning theories for teachers who want to support their students' learning theories, there are additional theories they may want to understand. Some may work as good supplemental theories for teachers to bolster their educational support with a new approach.

Connectivism

Connectivism is one of the newest learning theories. It focuses on the idea that people learn and grow when they form connections. This can be connections with each other or connections with their roles and obligations in their lives. Hobbies, goals, and people can all be connections that influence learning.

Transformative Learning Theory

Transformative learning theory is a great approach for adult education and young adult learning. Also referred to as transformation learning, transformative learning theory focuses on the idea that learners can adjust their thinking based on new information.

This learning theory was founded by **Jack Mezirow**, who discovered it after doing studies on adult women who went back to school. His initial research found that adults don't apply their old understanding to new situations and that having a new perspective helped them gain a new understanding of things as they change. Mezirow also believed that students had important teaching and learning opportunities connected to their past experiences and that critical reflection and review could lead to a transformation of their understanding.

This approach works well for adult students, as children don't have the same kind of transformation with their learning experiences—and with life experience. Adult students could draw on childhood experiences and transform those beliefs and understandings using critical reflection, leading them to an understanding of what they should believe and understand as adults.

Overall, the theory states that our worldview is changed the more we learn, which helps us grasp new concepts and ideas. By getting new information that helps evaluate past ideas, students are able to make a dramatic educational shift beyond standard learning. Teachers can employ this learning theory by encouraging their students to learn new perspectives while questioning their assumptions and open the floor for discourse to cement their new train of thought.

Social Learning Theory

Using social learning theory can be a valuable tool for dealing with difficult students who like to disrupt the classroom and cause trouble. This theory focuses on the concept of children learning from observing others by acting on or not acting on what they see exhibited by their classmates. For example, they may see a classmate politely asking for a treat and getting one, or maybe they hear another classmate talking about something new they've learned, which teaches the student something new even if it's not something they try themselves.

This learning theory was founded by **Albert Bandura**. He conducted an experiment called the *Bobo doll experiment* in the early '60s, during which he studied children's behaviour after they watched an adult act aggressively with a doll-like toy. He noted how the children reacted when the adult got rewarded, punished, or suffered no consequences after they attacked the

doll. Bandura wrote about his findings in 1977, detailing social learning theory and how it affected the behavioural development of students.

There are four elements to social learning theory:

- Attention, which calls upon different or unique lessons or activities to help children focus.
- Retention, focusing on how the student will internalise information and recall it later on.
- Reproduction, drawing on previously learned behaviour and when it's appropriate to use it.
- Motivation, which can extend from seeing other classmates being rewarded or punished for their actions.

By using social modelling based on these elements, teachers have a very powerful tool in their arsenal that can effectively guide their students to be more active in their learning, pay more attention, and channel their energy into their schooling.

Experiential Learning Theory

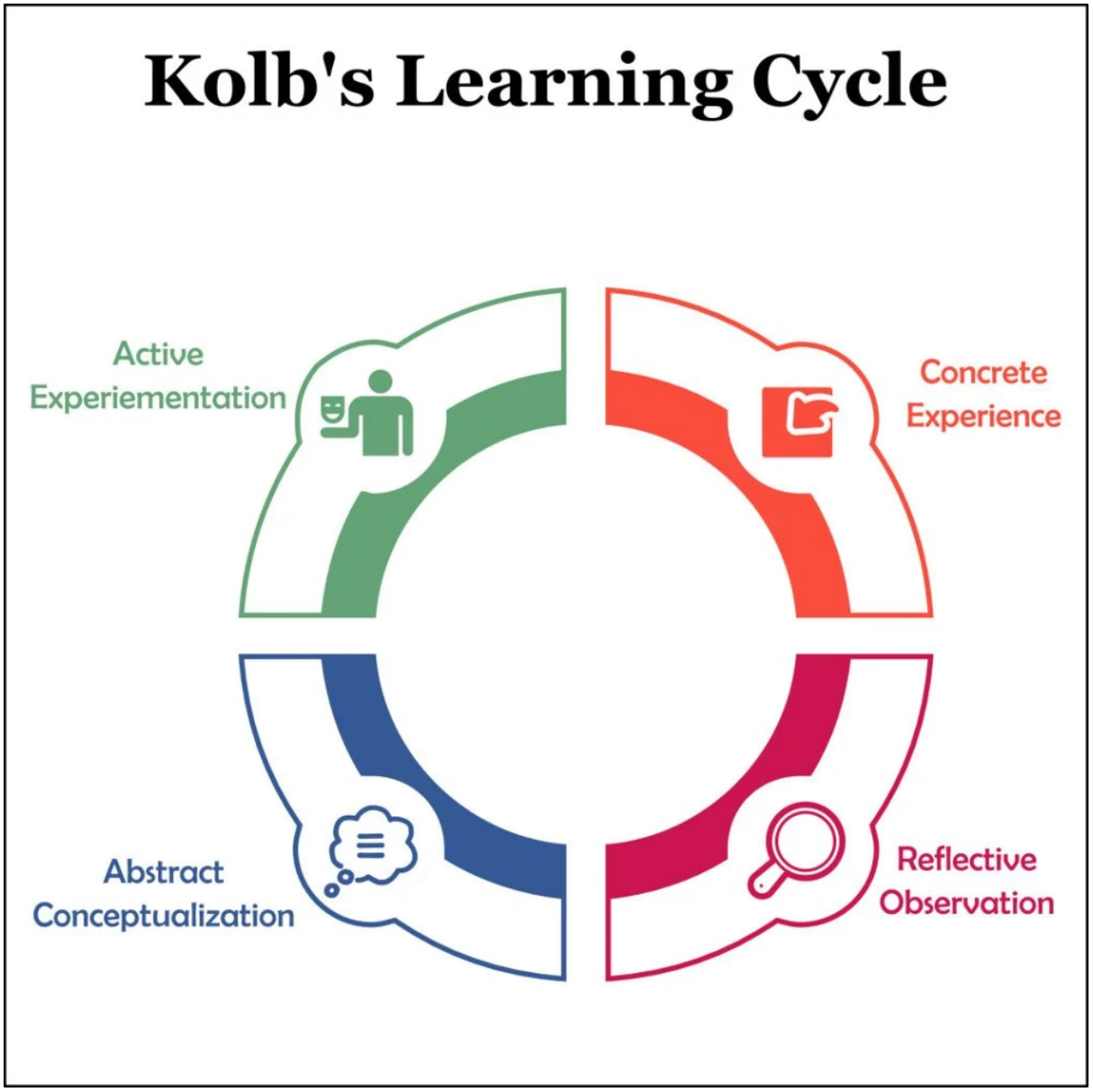
Experiential learning theory focuses on learning by doing. Using this theory, students are encouraged to learn through experiences that can help them retain information and recall facts.

Experiential learning theory, or ELT, was identified by David Kolb in 1984. Though his influence came from other theorists such as John Dewey, Kurt Lewin, and Jean Piaget, Kolb was able to identify four stages of ELT:

- Concrete Learning
- Reflective Observation
- Abstract Conceptualisation
- Active Experimentation

The first two stages, concrete learning and reflective observation, focus on grasping an experience. The latter two, abstract conceptualization and active experimentation are about transforming an experience. To Kolb, effective learning is seen as the learner goes through the cycle of

experiential learning theory. Students can enter the cycle in any way and at any point.



Some examples of this form of learning include taking students to the zoo to learn about animals instead of just reading about them or growing a garden to learn about photosynthesis instead of watching a video about it. By creating environments where students can learn and experience at the same time, teachers offer students the opportunity to immediately apply their knowledge and get real-world experiences. This approach also encourages teamwork and is shown to improve motivation.

Applying Learning Theories

Teachers can create specific strategies and techniques to apply these learning theories in their classrooms. Teachers need to first focus on getting a well-rounded education to learn about all kinds of techniques for teaching and classroom management. Teachers need to understand learning theories to be prepared to utilise them in their classrooms.

An understanding of learning theories helps teachers connect to all different kinds of students. Teachers can focus on different learning styles to reach different students, creating teaching that focuses directly on student needs and aptitudes.

Guide Student Learning through Enforcement

While the theory of behaviourism is no longer commonly used outside of the psychomotor domain, it emphasises guiding learners' behaviours through reinforcement. You can help achieve this by creating clean, consistent course design and including formative assessments that provide regular feedback.

To help create a clean, consistent course design that promotes accessibility and usability, follow these best practices:

- ➢ Map your content into instructional units (e.g., Weeks, Modules, Topics).
- ➢ Include measurable student learning objectives at the beginning of each instructional unit.
- ➢ Create consistent pages so that students become accustomed to where information is located. This will allow them to spend less time on trying to navigate the course, and more time on immersing themselves in the learning process.
- ➢ Work with an instructional designer.

In addition to course design, formative assessments are low-stake activities that guide students through the learning process. Formative assessments also give more opportunities to provide feedback for learning.

Make Learning Meaningful and Relevant

The theory of cognitivism focuses on making learning meaningful by using learners' background knowledge and experiences. Creating authentic assessments that require learners to reflect on and apply what they know will foster a meaningful connection to the material. You can also make learning meaningful through the following practices:

- ➢ Ask meaningful questions that focus on the deeper meaning instead of the minor details.
- ➢ Give students opportunities to collaborate and learn from each other.
- ➢ Create meaningful activities that give students the opportunity to apply new knowledge.
- ➢ Create multiple ways of interacting with students, and be available to guide and assist as students work through the coursework.

Cognitivists also stress that learners grasp more information at the beginning and end of content; when you build instructional materials, include "must-know" concepts at the beginning and the end.

Andragogy, the art and science of helping adults learn, also emphasises making learning relevant, meaningful, and practical. Because of outside commitments, adults tend to be more goal-oriented and motivated by practical application. Consider the following as you outline the instructional

materials and assessments to help meet intrinsic motivators for adult learners:

- ➢ Give students opportunities to highlight their prior experiences in new learning situations (e.g., case studies).
- ➢ Focus on goals, practicality, and relevancy (e.g., learning objectives, authentic assessments).
- ➢ Include elements in your course design and assessments that display respect for adults' previous experiences and responsibilities (e.g., self-directed learning).

Foster Self-Directed Learning

Both constructivism and andragogy emphasise the importance of self-directed learning, in which learners construct knowledge based on experience. Despite what its name suggests, self-directed learning does not have to be practised on an individual basis. You can also encourage self-directed learning in group settings that promote learning through experience and diverse perspectives. To foster self-directed learning:

- ➢ Incorporate learners' prior knowledge and interests.
- ➢ Provide tools for learners to succeed, but encourage independent responsibility.
- ➢ Promote experiential learning by including reflections.
- ➢ Create clear course goals, learning objectives, and assessment instructions to guide students through the learning process.
- ➢ Assume the role of "facilitator" instead of "sage on stage."

Importance

Why are learning theories important, and what value do they hold for educators and students?

Teachers make use of learning theories (consciously or not) and their related pedagogic approaches and technologies to design, develop and deliver effective teaching and learning experiences. But when we design consciously from learning theories, the overall effectiveness of the teacher-student dynamic increases noticeably. Understanding learning theories will help you deliver information effectively.

According to Pamela Roggeman, Dean of University of Phoenix's College of Education, the effects of applied educational theory can be long-lasting. She explains,

"The learning theories we experienced as a student influence the type of work environment we prefer as adults. For example, if one experienced classrooms based heavily on social learning during the K-12 years, as an adult, one may be very comfortable in a highly collaborative work environment. Reflection on one's own educational history might serve as an insightful tool as to one's own fulfilment in the workplace as an adult."

Put simply, learning theories provide actionable insight into how learning occurs — insight which, ideally, should be used to drive and inform each educator's approach to teaching and interacting with their students. Here are a few of ways that you and your students can benefit when you bring learning theories into your classroom or curriculum:

- Theories of learning can provide clarity and direction by offering a set of principles or guidelines to build your teaching approach around.
- The more theories of learning you're familiar with, the more strategies you'll have to connect with a diverse range of students who come from different backgrounds, learn at different paces, and face different academic challenges or obstacles.
- Learning theories facilitate clear communication between teachers and students (including nonverbal communication like body language, as we saw in the section on Behaviorism), along with parents, families, and school administrators.
- As learning outcomes improve, students will also build confidence and self-esteem.

Applying learning theories appropriately will help teachers do their jobs more effectively, so that students can achieve better outcomes. It is essential to have a concrete understanding of these theories and use them properly.

References

1) Learning theories: Definition - Learningbp
https://www.learningbp.com/learning-theories-definition-and-characteristics-every-educator-should-know/

2) Learning theory (education) - Wikipedia
https://en.wikipedia.org/wiki/Learning_theory_(education)

3) Learning Theories: Theories of Learning in Education | NU
https://www.nu.edu/blog/theories-of-learning/

4) Learning Theories - Leader In Me
https://www.leaderinme.org/blog/learning-theories/

5) Five Educational Learning Theories
https://www.wgu.edu/blog/five-educational-learning-theories2005.html#close

6) Applying Learning Theories - University of Florida
https://citt.ufl.edu/resources/the-learning-process/applying-learning-theories/

7) Learning Theory Overview | Instructional Design Central (IDC)
https://www.instructionaldesigncentral.com/learning-theory

8) What Is Humanistic Learning Theory in Education?
https://www.wgu.edu/blog/what-humanistic-learning-theory-education2007.html

9) Your Guide to Learning Theories (and Why They're Important)
https://www.goskills.com/Resources/Learning-theories

10) 5 Educational Learning Theories and How To Apply Them | UOPX
https://www.phoenix.edu/blog/educational-learning-theories.html